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Code Diversity in Model Training: A Comparative Study for Dataset Accuracy Enhancement in Plant Disease Detection.

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Corresponding Author:	Sunishtha Nangal, B.Tech Indira Gandhi Delhi Technical University for Women NEW DELHI, New Delhi INDIA
Corresponding Author Secondary Information:	
Corresponding Author's Institution:	Indira Gandhi Delhi Technical University for Women
Corresponding Author's Secondary Institution:	
First Author:	Ravinder M.
First Author Secondary Information:	
Order of Authors:	Ravinder M.
	Priyanka Kalra, B.Tech
	Sunishtha Nangal, B.Tech
	Jafrin Khan, B.Tech
	Yashica Dhalla, B.Tech
Order of Authors Secondary Information:	
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Abstract:	Plant disease detection is a major aspect of ensuring agricultural productivity and sustainability. Using Convolutional Neural Networks (CNNs) through deep learning techniques offers a promising method for automating this task. However, the efficacy of such models heavily depends on the datasets used for training, specifically the dataset quality. This paper investigates the influence of dataset quality on model performance and explores the potential of transfer learning approaches in addressing dataset limitations. Drawing from empirical studies, we analyse the effectiveness of CNNs and the utilisation of transfer learning strategies in plant disease detection, as demonstrated through the implementation of self-distillation and knowledge-distillation techniques. Furthermore, we address ongoing challenges in plant disease detection, including dataset scarcity and quality issues. By employing self-distillation and knowledge-distillation methods, we aim to refine model architectures and improve their ability to generalise across diverse datasets. Our findings emphasise the importance of leveraging transfer learning techniques to reduce the impact of dataset limitations and enhance the robustness of disease detection models. In conclusion, this paper underscores the significance of dataset quality and transfer learning approaches in advancing plant disease detection systems. By refining pre-processing algorithms and integrating sophisticated deep learning techniques, our goal is to enhance the accuracy and reliability of automated disease diagnosis, thereby contributing to agricultural sustainability and global food security initiatives.

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Code Diversity in Model Training: A Comparative Study for Dataset Accuracy Enhancement in Plant Disease Detection.

Ravinder M.[1], Priyanka Kalra[2], Sunishtha Nangal[3], Jafrin Khan[4], Yashica Dhalla[5]
[1] Assistant Professor, [1],[2],[3],[4],[5] Department of Computer Science and Engineering, Indira Gandhi Delhi Technical University for Women, New Delhi 110006, India
{ ravinderm@igdtuw.ac.in, priyankakalra615@gmail.com, sunishthanangal2001@gmail.com, khan.jafrin.9346@gmail.com, yashica.dhalla121@gmail.com }

ABSTRACT

Plant disease detection is a major aspect of ensuring agricultural productivity and sustainability. Using Convolutional Neural Networks (CNNs) through deep learning techniques offers a promising method for automating this task. However, the efficacy of such models heavily depends on the datasets used for training, specifically the dataset quality. This paper investigates the influence of dataset quality on model performance and explores the potential of transfer learning approaches in addressing dataset limitations. Drawing from empirical studies, we analyse the effectiveness of CNNs and the utilisation of transfer learning strategies in plant disease detection, as demonstrated through the implementation of self-distillation and knowledge-distillation techniques. Furthermore, we address ongoing challenges in plant disease detection, including dataset scarcity and quality issues. By employing self-distillation and knowledge-distillation methods, we aim to refine model architectures and improve their ability to generalise across diverse datasets. Our findings emphasise the importance of leveraging transfer learning techniques to reduce the impact of dataset limitations and enhance the robustness of disease detection models. In conclusion, this paper underscores the significance of dataset quality and transfer learning approaches in advancing plant disease detection systems. By refining preprocessing algorithms and integrating sophisticated deep learning techniques, our goal is to enhance the accuracy and reliability of automated disease diagnosis, thereby contributing to agricultural sustainability and global food security initiatives.

INTRODUCTION

Over the recent years, the emergence of deep learning methodologies and agricultural practices in collaboration have garnered considerable attention, particularly in the domain of plant disease detection. Agriculture stands as a linchpin of global food security, yet its productivity faces incessant threats from an array of biotic and abiotic stressors, with plant diseases looming as a primary menace. Traditionally, the identification and management of these diseases have heavily relied on manual inspection by trained agronomists, a labour-intensive and time-consuming endeavour that often culminates in delayed intervention and substantial crop losses. However, the emergence of notable deep learning techniques, in particular Convolutional Neural Networks (CNNs), has revolutionised this landscape by furnishing automated and precise solutions for plant disease detection through image analysis.

The application of CNNs in plant disease detection capitalises on its inherent capability to identify subtle features and features from image data, thereby enabling the identification of diseased plants with remarkable precision. This approach has not only streamlined the detection process but has also facilitated early intervention, thus curtailing yield losses and optimising resource allocation in agriculture. Moreover, the abundance expansive datasets such as the PlantVillage dataset, Agriculture-Vision dataset, and Plant Pathology Challenge 2020 dataset has provided researchers with a diverse array of labelled images encompassing various crops and diseases, fostering the evaluation and development of robust deep learning models.

Despite the promising advancements engendered by deep learning methodologies in plant disease detection, a myriad of challenges persists within this domain. Foremost among these challenges is the scarcity of high-quality and diverse datasets. While repositories like the PlantVillage dataset have served as invaluable resources for training deep

learning models, they often suffer from limitations such as inconsistent labelling, class imbalance, and diversity image quality. These shortcomings can encumber the generalisation prowess of deep learning models, hampering their performance in real-world contexts characterised by environmental heterogeneity. Another significant hurdle lies in the robustness of deep learning models across diverse environmental conditions. Models trained on specific datasets may exhibit suboptimal performance when confronted with novel environments or crops, thereby undermining their efficacy and reliability in practical applications. Additionally, the interpretability of deep learning models poses a conundrum, as their complex architectures obfuscate the underlying decision-making process, curtailing their transparency and trustworthiness in critical decision-making scenarios.

Against this backdrop, we venture on an extensive journey of deep learning techniques for plant disease detection, drawing insights from an exhaustive review of existing literature and the analysis of diverse implementation approaches. By synthesising key findings from multiple research papers and distilling insights from diverse methodologies, this paper endeavours to furnish a comprehensive understanding of the current landscape, identify critical challenges and limitations, and propose future directions to propel the field of plant disease detection forward using deep learning.

Beyond using deep learning models for plant disease detection, recent studies have delved into model distillation techniques to boost the efficacy and efficiency of these models. Model distillation, or knowledge distillation, entails transferring expertise from a sophisticated, high-capacity teacher model to a simpler, more compact student model. By distilling insights from the teacher model, the student model can achieve comparable performance with fewer computational demands, making it particularly well-suited for deployment on embedded systems.

By synthesising insights from existing literature, analysing various implementation strategies, and identifying key challenges and opportunities, this paper aims to achieve the following objectives:

1. **Assess the Effectiveness of Deep Learning Models:** Evaluate the performance of deep learning models, including CNNs, in plant disease detection tasks using diverse datasets such as the PlantVillage dataset, the Agriculture-Vision dataset, and the Plant Pathology Challenge 2020 dataset.
2. **Investigate Model Distillation Techniques:** Explore the application of model distillation techniques in enhancing the efficiency and scalability of plant disease detection via deep learning models. This involves examining the effectiveness of distillation in transferring knowledge from complex teacher models to simpler student models, as well as its impact on model performance and computational efficiency.
3. **Address Challenges and Limitations:** Identify and discuss the challenges and limitations inherent in current deep learning approaches for plant disease detection, including challenges associated with dataset quality, model robustness, and real-world deployment. Propose possible solutions and avenues for research to mitigate these challenges and advance the field.
4. **Propose Future Directions:** Outline future research directions and innovative areas in the domain of plant disease detection utilising deep learning including the integration of non-intrusive technologies, refinement of preprocessing algorithms, and exploration of interdisciplinary collaborations between computer science and plant pathology experts.

MATERIALS AND METHODS

Dataset

The PlantVillage dataset serves as a fundamental component of our research, providing a diverse collection of RGB images featuring healthy and unhealthy plant leaves across various crops. Comprising 87,000 images distributed across 38 classes, this dataset encompasses 14 crop species afflicted with 26 distinct diseases. The images exhibit a range of conditions, including variations in illumination, angles, surfaces, and noise levels, simulating real-world scenarios encountered in agricultural settings. Such diversity ensures the robustness and generalizability of our models for plant disease detection tasks.

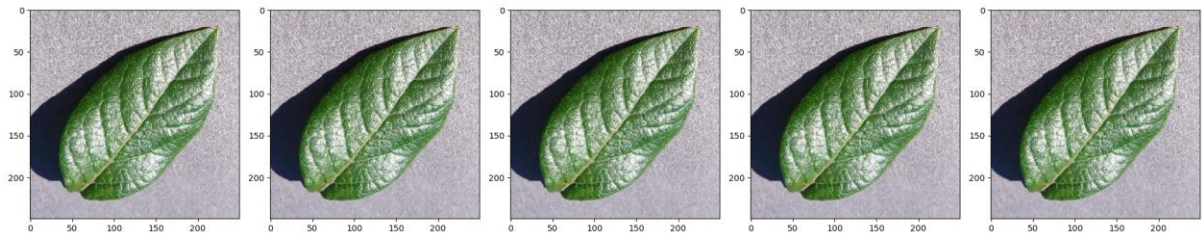


Fig. 1. Examples of the Plant Village dataset's augmented images.

Knowledge Distillation

Knowledge distillation utilising PlantVillage for the detection of plant diseases involves knowledge transferring from the complex model to a simpler model. The complex teacher model, which is often a CNN model, has been trained on the large and diverse PlantVillage dataset to accurately classify plant diseases. Through distillation, this knowledge is compressed into a more lightweight student model, enabling efficient deploying on resource-constrained devices like smartphones or embedded systems. By distilling the insights gained from the teacher model, the simpler model achieves high performance with significantly reduced computational resources making it ideal for real-time plant disease detection in the field applications.

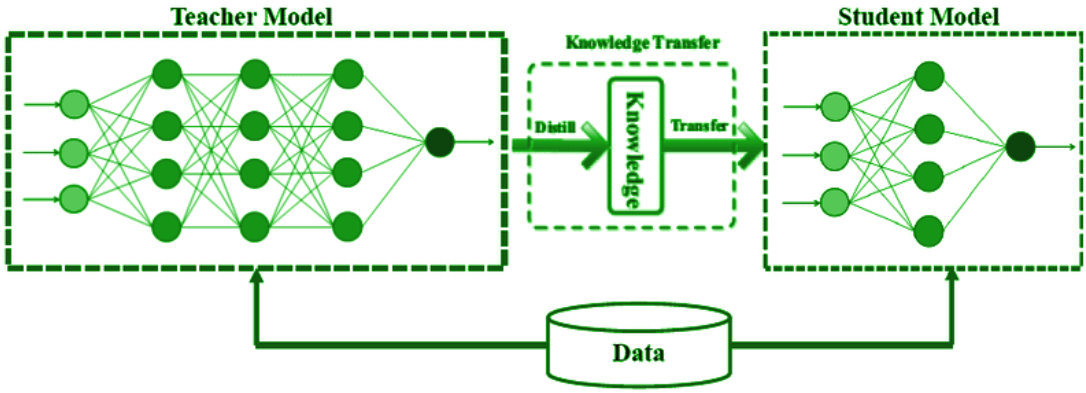


Fig. 2. Generic architecture of knowledge distillation using a teacher-student.

Teacher Model

The teacher model is typically complex and a powerful neural network, often based on deep learning architectures such as CNNs. It is trained on a comprehensive and varied dataset, like the PlantVillage dataset, encompassing images depicting diverse plant diseases and healthy plants. During training, the teacher model acquires the knowledge to classify images accurately into different disease categories or label them as healthy. The teacher model's primary objective is to achieve high accuracy on the training dataset, capturing complex features and patterns present in the images.

Student Model

The student model is a streamlined version of the intensive teacher model, which is specifically designed to be computationally lightweight and efficient. It typically has a reduced number of parameters and requires less computational requirements compared to the teacher model. The student model's architecture may be simpler, such as a shallower CNN or even a different type of model altogether, like a decision tree or a linear classifier.

Since the simpler model learns from the teacher model's predictions rather than directly from the training data, it aims to replicate the teacher model's behaviour by capturing the essential knowledge encoded in its predictions.

Despite its simplicity, the student model intends to achieve performance comparable to the teacher model, providing accurate predictions on unseen data while being efficient enough for real-time application.

EVALUATION METHODS

METHODOLOGY 1

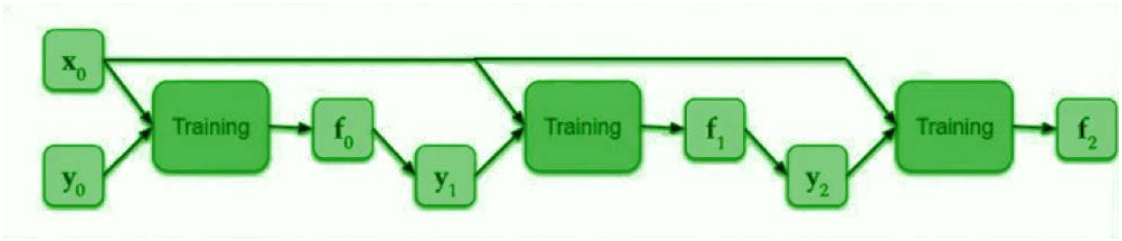


Fig. 3. Generic architecture of self-knowledge distillation using a single model.

We commence our research by adopting the self-distillation approach, recognizing its suitability for grasping the core principles of model distillation while maintaining simplicity. Self-distillation facilitates continual learning within a single model by gradually refining the student model's predictions over multiple training epochs, eliminating the need for external teacher models or additional data sources. This methodology enables the student model to refine its predictions based on its own past outputs, promoting performance improvement and generalisation over time.

Student Model Architecture and Training

Our student model is designed as a compact architecture comprising 2 convolutional layers for extracting features from the input images, 2 LeakyReLU activation functions, 2 max-pooling layers for downsampling the spatial dimensions and one dense output layer for classification. The model was constructed to process input images of size 250x250x3 and output class probabilities corresponding to the various diseases affecting plants in the PlantVillage dataset. It underwent training with augmented data, categorical cross-entropy loss function, and the Adam optimizer using the fit method for 5 epochs while monitoring its performance on the validation data.

Layers	Student Model
Conv2D	2
LeakyReLU	2
Max Pooling	2
Flatten	1
Dense	1
Input	1
Output	1
Types of Layers	7
Trainable Layers	5
Total Layers	10

Table 1.1. Layers analysis for implementation 1.

Adam Optimiser Algorithm:

The Adam optimiser update rule is expressed as follows:

$$m_t = \beta_1 . m_{t-1} + (1 - \beta_1) . \nabla \boldsymbol{J}(\theta_t)$$

$$v_t = \beta_2 . m_{t-1} + (1 - \beta_2) . (\nabla \boldsymbol{J}(\theta_t))^2$$

$$\hat{m}_t = m_t \,/\, 1 - \beta_1^t$$

$$v_t = \hat{v}_t \,/\, 1 - \beta_2^t$$

$$\theta_{t+1} = \theta_t - \alpha . \hat{m}_t \,/\, (\hat{m}_t^{1/2} + \epsilon)$$

Where:

- $\boldsymbol{\theta_t}$ symbolises the model's parameters at the tth iteration.
- $\boldsymbol{\nabla J(\theta_t)}$ represents loss function $\boldsymbol{J}(\theta)$'s gradient with respect to the parameters θ, evaluated at θ_t .
- $\boldsymbol{\alpha}$ is for the learning rate, which determines the step size taken during optimization.
- $\boldsymbol{\beta_1}$ denotes the exponential decay rate for the first moment, while $\boldsymbol{\beta_2}$, the exponential decay rate for the second.
- $\boldsymbol{m_t}$ represents the first moment of the gradients, while $\boldsymbol{v_t}$ represents the second moment of the gradients.
- $\boldsymbol{e}$ is a minor constant offset added to the denominator to avoid division by zero, (e.g., 10^-6).
- $\boldsymbol{\hat{m}_t}$ and $\boldsymbol{\hat{v}_t}$ are bias-corrected estimates of the gradients' first and second moments, respectively.

Self-Distillation Process

Subsequently, we implement the self-distillation process, employing a custom distiller class to generate soft pseudo-labels and utilising a distillation loss function, the Kullback-Leibler Divergence. The 'Distiller' class was designed to take the student model as input and generate soft pseudo-labels from its predictions using a specified temperature. During training, the distillation loss was combined with the standard categorical cross-entropy loss using a predefined alpha weight to guide the optimization process.

The **distillation loss** is computed using the KLDivergence between the teacher model's **softened predictions** and student model's output. This formulation allows the student model to learn from both the true labels and the teacher's probabilistic predictions.

Distillation Loss = KLDivergence(q||p)

Where:

- q represents the output with softened probabilities by the teacher model.
- P represents the output with softened probabilities by the student model.
- The KL divergence quantifies the difference between a given probability distribution and a reference or expected distribution.

Softened Predictions = $softmax(\frac{logits}{T})$

Overall Loss = $\alpha \times Student\ Loss + (1-\alpha) \times Distillation\ Loss$

Where α represents the weight parameter regulating the balance between the student's loss and the distillation loss.

The self-distillation training was conducted using the 'fit' method on the augmented training data for 5 epochs, empowering the student model to refine its own predictions and boost its accuracy, which is chosen as the evaluation metric during training.

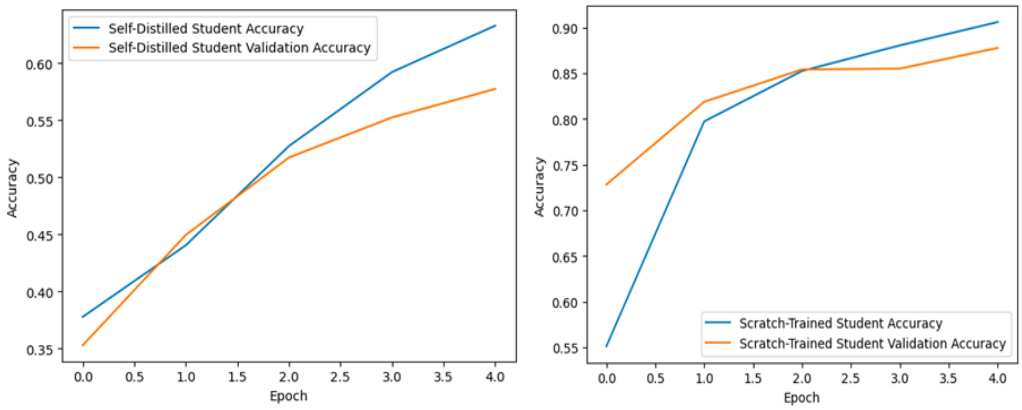
Training from Scratch for Comparison

For comparative analysis, we also trained a clone of the student model without employing self-distillation. This scratch-trained student model underwent similar compilation and training procedures as the self-distilled student model but without the distillation loss component.

Performance Evaluation and Visualization

To evaluate the effectiveness of self-distillation, we monitored and contrasted the training and validation accuracy of between the self-distilled student model and the scratch-trained student model across epochs. Line plots were

generated to visualise the accuracy trends, offering insights into the convergence behaviour and performance stability of the two approaches.



Graph Set 1. Trends observed for implementation 1.

Results

Overall, the architecture of the student model balances simplicity with effectiveness, enabling it to serve as both a teacher and a student in the knowledge distillation process for plant disease classification. The training and evaluation results provide valuable insights into the performance of the teacher, student (after distillation), and student (trained from scratch) models on the PlantVillage dataset.

Metric/Model	Teacher Model	Student Model (After Distillation)	Student Model (Trained from Scratch)
Architecture			
Total Layers	10	10	10
Types of Layers	7	7	7
Trainable Layers	5	5	5
Parameters			
Total params	12,50,310	12,50,310	12,50,310
Trainable params	12,50,310	12,50,310	12,50,310
Non-trainable params	0	0	0
Performance Metrics			
Training Accuracy	0.8967	0.6328	0.9059
Training Loss	0.3444	-	0.3221
Distillation Loss	-	1.74E-04	-
Validation Loss	0.4667	1.65E-04	0.4136
Validation Accuracy	0.8641	0.5774	0.8776

Table 1.2. Result analysis for implementation 1.

METHODOLOGY 2

As we observe, the self-distillation model didn't provide us with the satisfactory results expected. The most significant concern is the drop in both training and validation accuracy after the distillation. Self-distillation should ideally improve or at least maintain the performance of the student model, but in this instance, it deteriorated. Furthermore, the extremely low distillation loss (1.74E-04) suggests that the student model overfits the predictions by the teacher model on the training data which could be the cause for the drop in generalisation performance on the validation set. Hence, this leads us to explore an alternative distillation technique, Single Teacher-Student Distillation, that could be more effective for this dataset and the model architecture.

In our revised methodology, notable changes were made with respect to the first implementation:

- The addition of a teacher model allows a distillation process where the student acquires knowledge from the teacher's knowledge, which expectantly improves the student model's performance.
- The Distiller class was updated to incorporate the teacher model's predictions and the distillation loss. 1x1 convolutional layers were added for the teacher model while no 1x1 convolutions were for the student model.
- The training procedure now involves training the teacher model first and then using it to distil knowledge into the student model.
- The student loss function remains the same, but the distillation loss function was introduced as Kullback-Leibler Divergence (keras.losses.KLDivergence()). Metrics were updated to include both student and distillation losses.
- An alpha parameter was introduced to weigh the contribution of both the distillation loss and the student loss in the overall loss.

Teacher Model Architecture

The teacher model boasts a comprehensive architecture comprising five Conv2D layers, followed by a ReLU activation layer, and subsequent BatchNormalization layers applied to normalise the activations. Adding on, 3 MaxPooling2D layers and 4 Dropout layers act as a regularisation technique to prevent overfitting. The architecture also includes a flattened layer, followed by two Dense layers; for converting the 2D feature maps into 1 D feature vectors and classification purposes respectively. This teacher model uses an Adam optimizer as well, just like the student model in the previous distillation technique, while accuracy remains the evaluation metric.

Layer Type	Teacher Model	Student Model (After Distillation)	Student Model (Trained from Scratch)
Conv2D	5	2	2
Activation/LeakyReLU	6	2	2
Batch Normalization	6	-	-
MaxPooling2D	3	2	2
Dropout	4	-	-
Flatten	1	1	1
Dense	2	1	1
Input	1	1	1
Output	1	1	1

Types of Layers	9	7	7
Trainable Layers	15	5	5
Total Layers	29	10	10

Table 2.1. Layer analysis for implementation 2.

Student Model Architecture

When designing the student model, the goal shifts towards creating a more compact and efficient version of the teacher model without sacrificing too much in terms of performance. The architecture is simplified by reducing the number of layers and parameters while still aiming to capture the essential features learned by the teacher model.

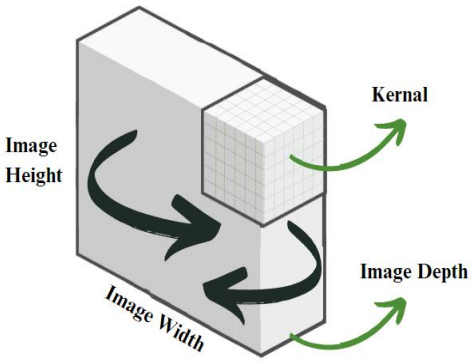


Fig. 4. 1 x 1 Convolutional filters.

As mentioned in the changes made, the teacher model has the 1x1 convolution layers, while the student model does not have any. This was done with the primary goal of ensuring the student model remains a lightweight version of the teacher model. Adding 1x1 convolutions would introduce additional complexity and parameters, which is counterproductive to this goal.

A simpler student model is less likely to overfit the teacher's predictions and can generalise better to unseen data, leading to improved validation accuracy and overall performance.

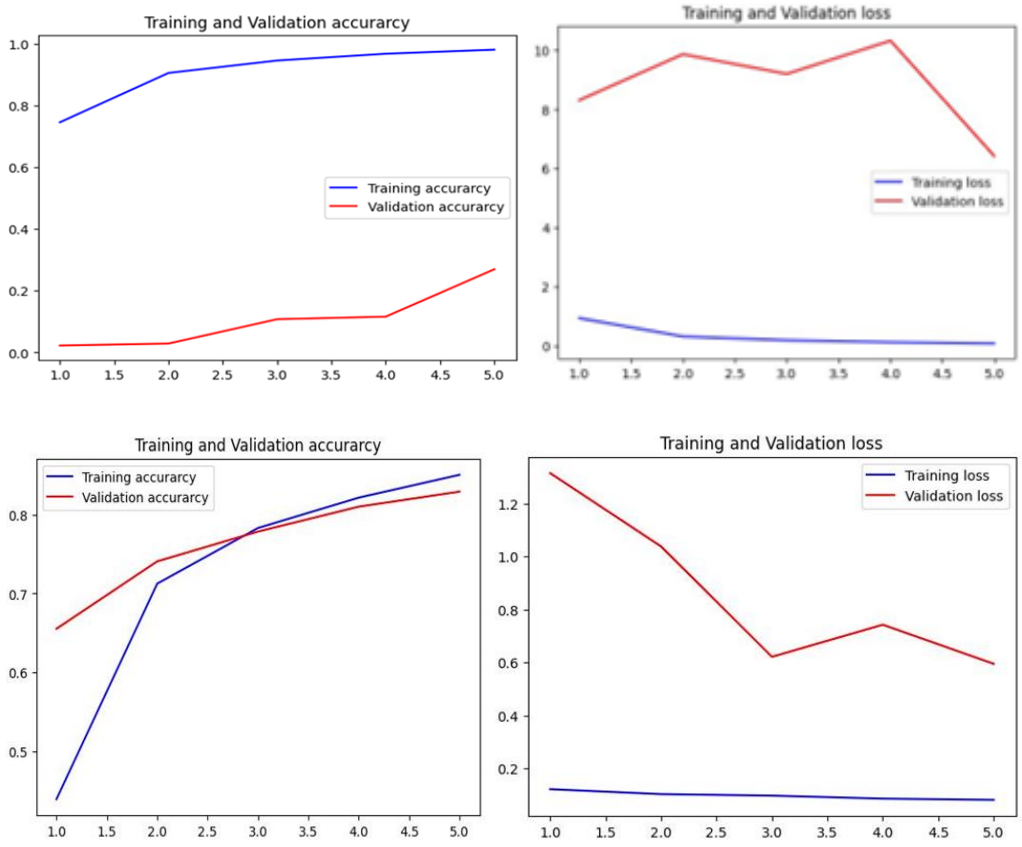
Additionally, it can efficiently learn from the teacher model's knowledge without getting overwhelmed by unnecessary complexity, facilitating better knowledge transfer during distillation. Requiring less computational resources, it is also faster to train, which is an aspect surely to be considered. Hence, the omission of 1x1 convolutions is a deliberate choice to focus on simplicity and efficiency.

Training from Scratch for Comparison

The scratch-trained student model underwent similar compilation and training procedures as the student model, but without the distillation loss component.

Performance Evaluation and Visualization

Accuracy of the distilled student model and the scratch-trained student model across epochs were checked and line plots were generated to visualise the training and validation accuracy trends.



Graph Set 2. Trends observed for implementation 2.

Results

Metric/Model	Teacher Model	Student Model (After Distillation)	Student Model (Trained from Scratch)
Architecture			
Total Layers	29	10	10
Types of Layers	9	7	7
Trainable Layers	15	5	5
Parameters			
Total Params	52,752,294	1,250,310	1,250,310
Trainable Params	52,749,414	1,250,310	1,250,310
Non-trainable Params	2,880	0	0
Performance Metrics			
Training Loss	0.065	0.6062	0.4537
Training Accuracy	0.9808	0.8504	0.8705
Distillation Loss	0	0.0828	-

Validation Loss	6.4153	0.7181	-
Validation Accuracy	0.2696	0.8291	-

Table 2.2. Layer analysis for implementation 2.

The teacher model, being more complex, has a high training accuracy but struggles with generalisation as seen from the low validation accuracy. This suggests overfitting on the training data.

The student model after distillation has improved in the metric of training accuracy compared to the model trained from scratch but falls behind in validation accuracy and training loss. This suggests that while the distillation has transferred some knowledge from the teacher to the student, it may not have been sufficient to capture the full complexity and nuances of the teacher model.

METHODOLOGY 3

The accuracy obtained in the previous implementation still isn't satisfactory. Teacher model has the highest training accuracy of 0.9808, followed by the scratch-trained student model with 0.8705, and then the student model after distillation with 0. 8504.

In response to the persisting unsatisfactory accuracy, we introduce modifications aiming to enhance the performance of the distilled student model. This involves increasing the dropout rates in the teacher model while maintaining the student model's architecture largely unchanged from previous iterations.

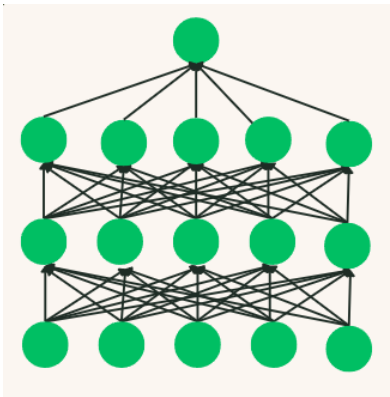


Fig. 5. Standard Neural Network.

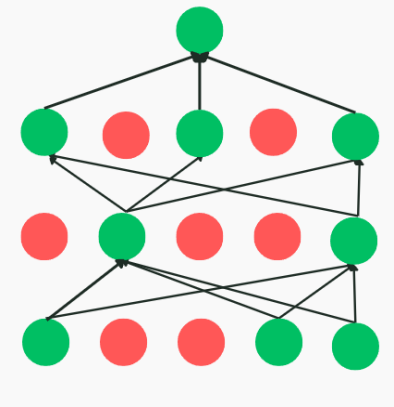


Fig. 6. After applying dropout rates

Dropout is a regularisation technique which is leveraged to prevent overfitting during training by randomly deactivating a fraction of input units, promoting the learning of more robust features, and reducing reliance on specific neurons. By augmenting dropout rates in the teacher model, we intensify regularisation to enhance its generalisation capability, thereby potentially improving the distilled student model's performance.

Teacher and Student Model Architectures

The decision to retain consistency in the student model's architecture across implementations allows for a clearer evaluation of the how alterations to the teacher model affect the performance of the distilled student model.

Layers	Teacher Model	Student Model (After Distillation)	Student Model (Trained from Scratch)
Conv2D	5	2	2

Activation	6	2	2
Batch Normalization	6	-	-
Max Pooling	3	2	2
Dropout	4	-	-
Flatten	1	1	1
Dense	2	1	1
Input	1	1	1
Output	1	1	1
Types of Layers	9	7	7
Trainable Layers	15	5	5
Total Layers	29	10	10

Table 3.1. Layer analysis for implementation 3.

Results

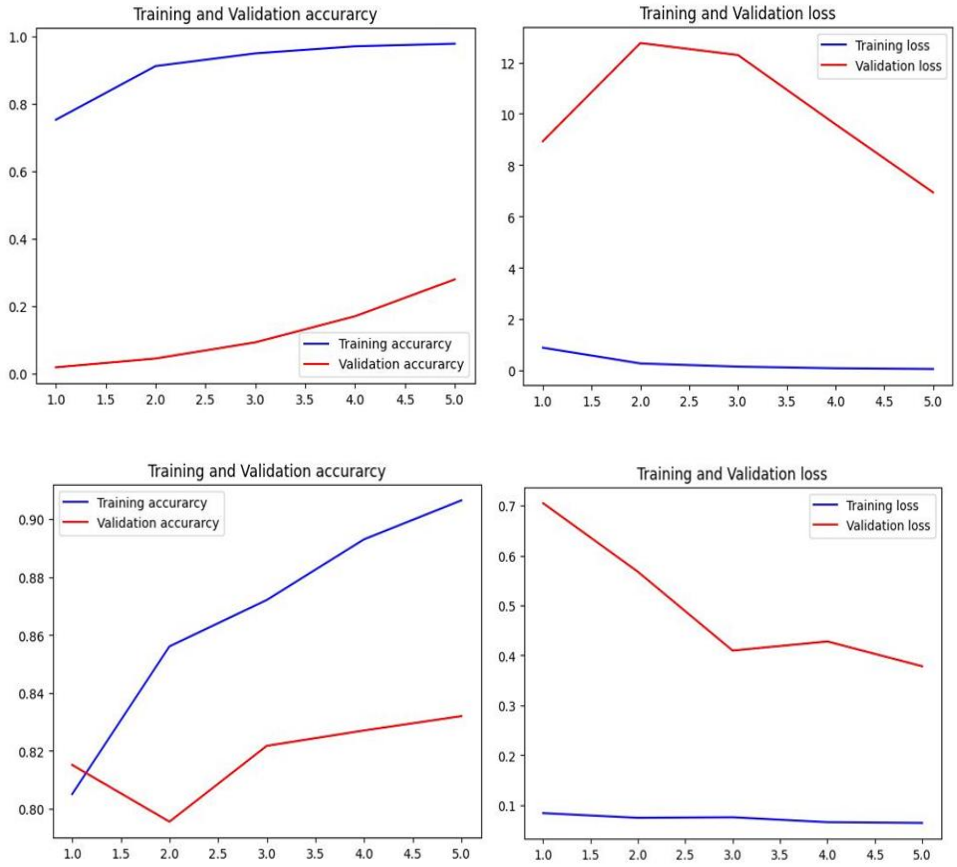
Our findings indicate considerable advancements in the performance of the distilled student model. These improvements highlight its ability to generalise better, as evidenced by lower validation loss and higher validation accuracy when compared to the scratch-trained student model.

The distilled student model effectively captures essential features learned by the teacher model, reflected in reduced distillation loss and improved performance metrics compared to previous approaches.

Metric/Model	Teacher Model	Student Model (After Distillation)	Student Model (Trained from Scratch)
Architecture			
Total Layers	29	10	10
Types of Layers	9	7	7
Trainable Layers	15	5	5
Parameters			
Total Parameters	5,27,52,294	12,50,310	12,50,310
Trainable Parameters	5,27,49,414	12,50,310	12,50,310
Non-trainable Parames	2,880	0	0
Performance Metrics			
Training Loss	0.1145	0.6231	1.0737

Training Accuracy	0.9643	0.8475	0.7053
Distillation Loss	-	0.1069	-
Validation Loss	10.7436	0.7776	-
Validation Accuracy	0.1105	0.8274	-

Table 3.2. Result analysis for implementation 3.



Graph Set 3. Trends observed for implementation 3.

METHODOLOGY 4

We finally achieved a higher accuracy with the distilled student model compared to the scratch-trained student model in the previous implementation. This helps us believe we're on the right track. One thing we still need to focus on though, is the gap between the distilled student model and the teacher model accuracies. The student model must be able to achieve a similar level of performance to the teacher model even with reduced complexity, for our research purpose to be successful.

Therefore, in our pursuit of enhancing the performance and bridging the gap between the distilled student and the teacher model model accuracies, significant modifications were implemented in the teacher model aimed to optimise its training dynamics and improve overall performance without altering its underlying feature extraction capabilities.

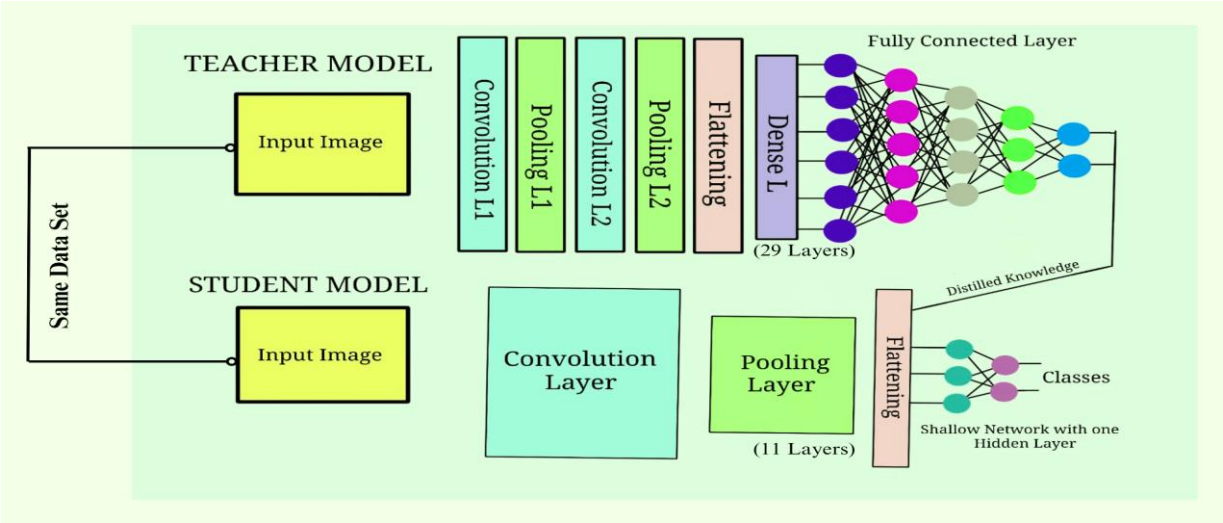


Fig. 7. Single teacher-student model knowledge distillation.

Teacher Model Architecture

Modifications made to the teacher model include the reduction in the dropout rates from 0.5 to 0.25 to minimise information loss during training, thereby improving the model's ability for generalisation of new data. In terms of distillation parameters, we increased the temperature from 3 to 10 to soften probability distributions, aiding the student model in capturing the teacher's knowledge more broadly. Additionally, we maintained the alpha value at 0.2, ensuring that most of the loss originates from the distillation loss rather than the student's predictions.

Student Model Architecture

Since the primary goal is to enhance the student model's performance with respect to accuracy and generalisation, a balance is required to ensure the student model learns from the teacher without deviating too much from the ground truth. The alpha parameter and the combined loss function facilitate this balance. By introducing specific loss functions and updating the Distiller class, the distillation process becomes more structured and effective which is likely to yield better results compared to ad-hoc methods.

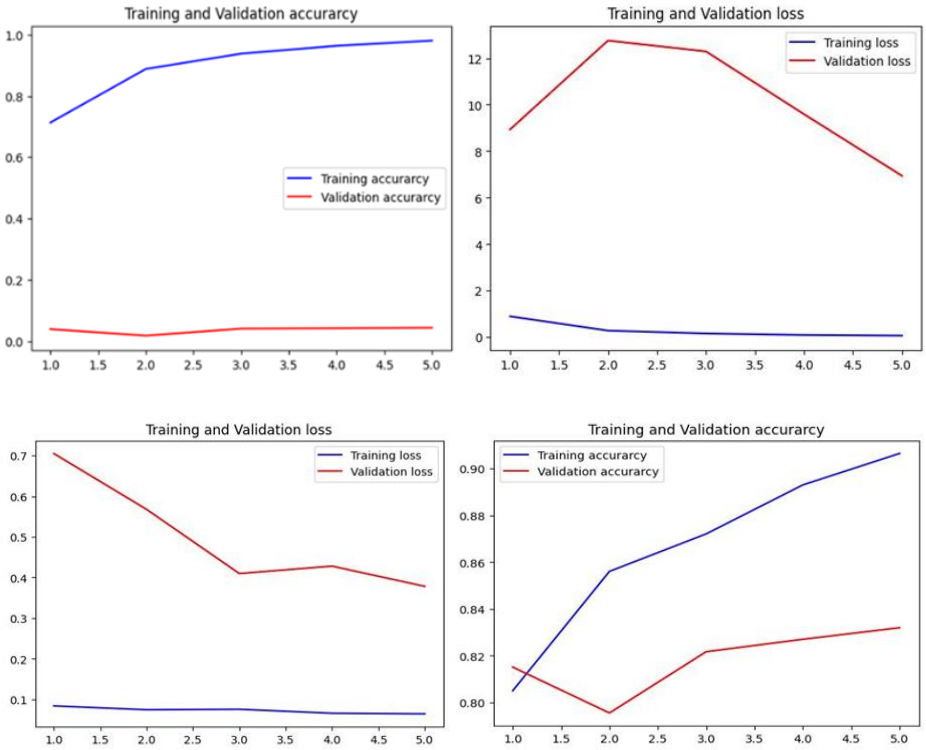
Layers	Teacher Model	Student Model (After Distillation)	Student Model (Trained from Scratch)
Conv2D	5	2	2
Activation/LeakyReLU	6	2	2
Batch Normalization	6	-	-
Max Pooling	3	2	2
Dropout	4	-	-
Flatten	1	1	1
Dense	2	2	2
Input	1	1	1

Output	1	1	1
Total Types	9	7	7
Trainable	15	5	5
Total Layers	29	11	11

Table 4.1. Layer analysis for implementation 4.

Evaluation

Both models are compiled with appropriate cross-entropy loss functions and evaluation metrics as accuracy to meet their training objectives. A student model trained from scratch serves as the comparison point to understand the changes in the accuracies achieved.



Graph Set 4. Trends observed for implementation 4.

Results

Metric/Model	Teacher Model	Student Model (After Distillation)	Student Model (Trained from Scratch)
Architecture			
Total Layers	23	11	11
Types of Layers	9 s	7	7
Trainable Layers	11	6	6

Parameters			
Total params	23,665,574	4,307,558	4,307,558
Trainable params	23,612,454	4,307,558	4,307,558
Non-trainable params	53,120	0	0
Performance Metrics			
Training Loss	0.0434	0.3108	0.6686
Training Accuracy	0.9858	0.9472	0.8054
Distillation Loss	-	0.0208	-
Validation Loss	0.6624	0.4934	-
Validation Accuracy	0.8315	0.8787	-

Table 4.2. Result analysis for implementation 4.

Post-distillation, the student model shows a distillation loss of 0.0691 and training accuracy of 90.65%, indicating successful knowledge transfer from the teacher. This results in improved validation performance for the student model with 83.20% accuracy and a loss of 0.6324, showcasing reduced overfitting and enhanced generalisation compared to the teacher.

While a student model trained from scratch achieves a training accuracy of 67.86%, slightly lower than the distilled student, its metrics are inferior overall, emphasising the benefits of knowledge distillation in improving model performance.

METHODOLOGY 5

In response to the undesirable widening gap between the accuracies of the distilled model and the teacher model observed in previous implementations, we opted to integrate ResNet50 as the teacher model in our distillation process, aiming to leverage its pre-trained features and regularisation capabilities.

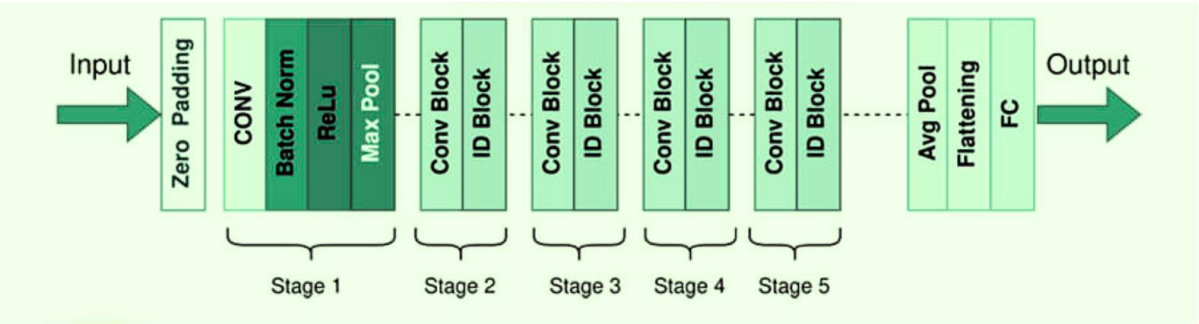


Fig. 8. Standard ResNet50 Architecture

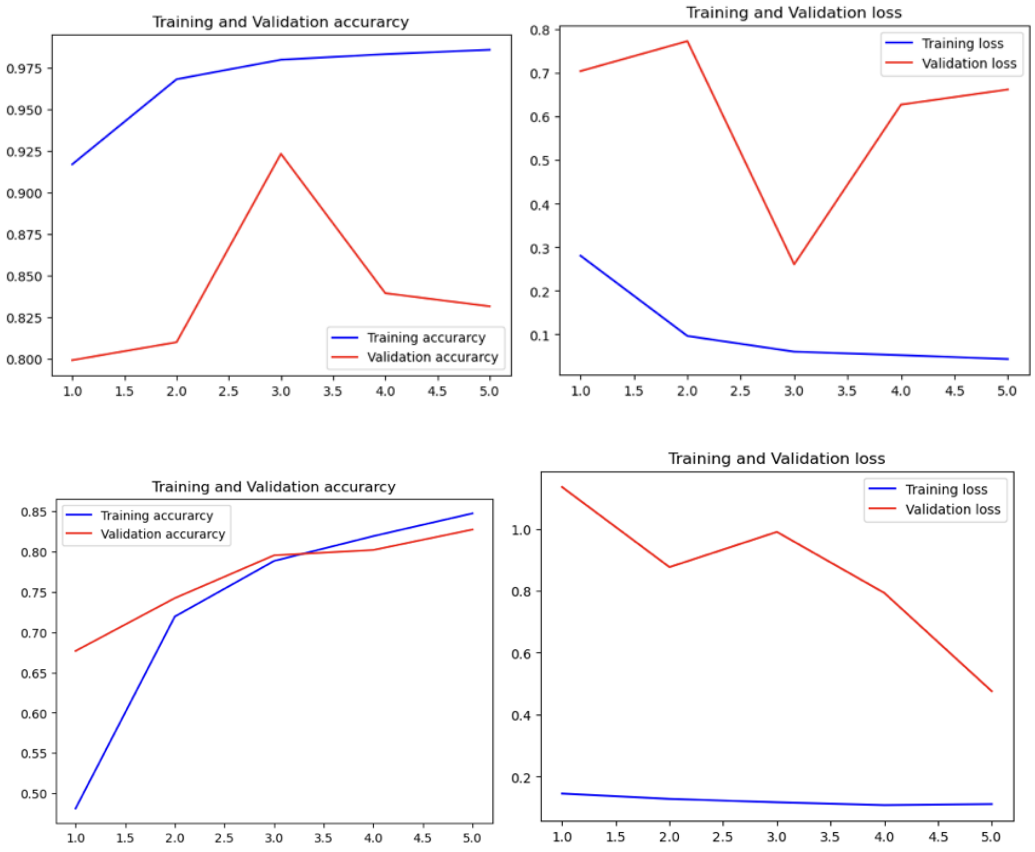
Residual Network, also known as ResNet50, is a convolutional neural network composed of several building blocks, each block containing multiple convolutional layers. With 50 layers, out of which one is a MaxPool layer, one is the average pool layer and the rest of 48 layers are the convolutional layers, it also includes skip connections (residual connections) which help in training deeper networks effectively.

Teacher and Student Model Architecture

Incorporating ResNet50 as the teacher model in our distillation process was an approach to address the gap in accuracies observed in our previous implementation. It brings the advantage of leveraging pre-trained features, depth, and regularisation capabilities to potentially enhance the performance and generalisation of our distilled student model. Meanwhile, the student architecture is majorly the same with only a decrease in the dropout layers to aim for better generalisation and avoid overfitting.

Metric/Model	Student Model (After Distillation)	Student Model (Trained from Scratch)
Conv2D	2	2
Activation/LeakyReLU	2	2
Batch Normalization	-	-
Max Pooling	2	2
Average Pool Layer	-	-
Dropout	-	-
Flatten	1	1
Dense	1	1
Input	1	1
Output	1	1
Total Types	6	6
Trainable Layers	5	5
Total Layers	10	10

Table 5.1. Layer analysis for implementation5..



Graph Set 5. Trends observed for implementation 5.

Result

Metric/Model	Teacher Model	Student Model (After Distillation)	Student Model (Trained from Scratch)
Architecture			
Total Layers		6	6
Total Types of Layers		5	5
Total Trainable Layers		10	10
Parameters			
Total params	23,665,574	4,307,558	4,307,558
Trainable params	23,612,454	4,307,558	4,307,558
Non-trainable params	53,120	0	0
Performance Metrics			
Training Loss	0.0434	0.3108	0.6356
Training Accuracy	0.9858	0.9472	0.8123
Distillation Loss	-	0.0208	-

Validation Loss	0.6624	0.4934	-
Validation Accuracy	0.8315	0.8787	-

Table 5.2. Result analysis for implementation 5.

This implementation is a great improvement in comparison to all the past implementations. The distilled student model finally achieved a satisfactory accuracy of 94.72%, surpassing previous benchmarks, exhibiting a reduced gap of 3.86% compared to the teacher model. It has a lower training loss, indicating better convergence and model fit to the training data. Even the distillation loss is much lower in comparison, indicating that the student model has successfully captured the knowledge from the teacher model. Lastly, lower validation loss and a higher validation accuracy point to better generalisation and performance on unseen data as well.

On comparing it with the scratch-trained student model, it is easy to gather that our distilled model is performing much better, hence giving an insight into the power of distillation Thus, our refined distillation approach successfully met our initial goals of achieving comparable accuracies and improving accuracy through distillation.

RESULT ANALYSIS

Experimental details

For the experimental setup, we developed the model using TensorFlow for both training and testing, running it on an 8-core GPU. The training dataset includes 43,429 samples and the validation dataset includes 5,459 samples from the PlantVillage [35] dataset. During training, we optimised the model using Adam optimizer with epochs set to 5 and batch size set to 500. To optimise the performance of the models we have used Image Data Generator to augment and preprocess the images prior to providing them into the model. The rescale parameter was set to 1.0/255 for both training and validation data generators. This operation standardises pixel values to the range [0, 1], a standard preprocessing step for neural network models. The use of data generators allows for real-time data augmentation, which helps improve model generalisation and performance. By rescaling pixel values to the range [0, 1], numerical stability is improved, and the model is better able to learn from the data. This step is crucial in preparing the data for training and validation, ensuring that the model receives properly pre-processed and augmented inputs.

Performance Comparison

Across five methodologies aimed at enhancing accuracies via model distillation on the PlantVillage Dataset, various strategies were explored and refined. Initially, self-distillation was employed, allowing the student model to distil knowledge from its own predictions, but the results were not satisfactory, indicating room for improvement. Subsequently, the focus shifted towards single teacher-student distillation, where knowledge is transferred from a larger teacher model to a smaller simpler student model. However, although this approach exhibited promise, there was still a disparity between the accuracies of the teacher and distilled student models. Specifically, the student model after distillation achieved a training accuracy of 67.40% and a distillation loss of 0.4523. This gap suggested the need for further optimization, particularly to bridge the performance differential between the teacher and student models.

Recognizing the necessity for refinement, subsequent methodologies introduced significant modifications to both teacher and student model architectures, aiming to bridge the gap in accuracy. Augmenting dropout rates in the teacher model aimed at enhancing generalisation, while alterations to layer configurations and the introduction of

new regularisation techniques sought to improve the distilled student model's performance. However, while these efforts yielded improvements, there was still a gap in accuracy between the student and teacher models. For example, the distilled student model showcased better generalisation with a validation accuracy of 64.41% and a validation loss of 1.6524 compared to the scratch-trained student model. Nonetheless, this gap underscored the complexity of the distillation process and the challenge of achieving comparable performance in smaller models.

The integration of ResNet50 as the teacher model marked a pivotal shift, leveraging pre-trained features and regularisation capabilities to enhance knowledge transfer. Simplifications to the student model architecture aimed at better generalisation, resulting in a notable improvement in the distilled student model's validation accuracy, of 87.87% and a validation loss of 0.4934, showcasing a reduced gap compared to the teacher model. This final implementation showcased the power of leveraging pre-trained models and strategic architectural adjustments in achieving satisfactory accuracies through distillation. With a training accuracy of 94.72%, the distilled student model surpassed previous benchmarks, highlighting the importance of continual refinement and innovation in model distillation techniques to meet research objectives effectively.

Ablation study

In conducting an ablation study, the comparison between distilled student models and those trained without knowledge of distillation reveals intriguing insights. While distilled student models may not consistently outperform their non-distilled counterparts in terms of Frames Per Second (FPS) on GPU and CPU, they exhibit notably superior performance in mean Average Precision (mAP) at different Intersections over Union (IoU) thresholds. These models demonstrate marked improvements across key evaluation metrics, showcasing higher mAP scores at varying IoU thresholds while maintaining competitive FPS rates on both GPU and CPU. This suggests that although non-distilled student models might achieve faster inference speeds, they struggle to achieve exceptional performance levels in tasks such as plant disease detection, where high accuracy is paramount.

The findings underscore the trade-off between accuracy and speed inherent in the training process. While non-distilled student models may excel in computational efficiency, their performance in critical metrics like mAP is comparatively lacklustre. On the contrary, distilled student models strike a more optimal balance between speed and accuracy, making them better suited for tasks requiring precise detection, such as plant disease identification. Consequently, these results highlight the nuanced considerations that practitioners must weigh when selecting between distilled and undistilled approaches, emphasising the importance of aligning model architecture and training strategies with the specific requirements and constraints of the application domain.

Evaluation model lightweight

In evaluating models for real-time applications, the focus lies on achieving a balance between computational efficiency and predictive performance. Each model variant undergoes scrutiny based on key metrics such as the number of layers, parameters, evaluation loss, and accuracy to ascertain its applicability for deployment in real-world scenarios
In our quest to optimise model performance and bridge the accuracy gap between the teacher model and the distilled student model, we initiated a series of methodological refinements.

Initially, we embraced self-distillation as our inaugural approach, a technique known for its simplicity and potential for continual learning within a single model. By empowering the student model to distil knowledge from its own past predictions, we anticipated performance enhancements and improved generalisation over time. However, our endeavours with self-distillation encountered setbacks, with both training and validation accuracies diminishing post-distillation. This unexpected outcome prompted a strategic pivot towards the single Teacher-Student Distillation methodology. Here, we transferred insights from a larger, more complex teacher model to a streamlined student counterpart, leveraging a distillation process to facilitate knowledge transfer. Notably, we omitted 1x1 convolutional

layers in the student model to prioritise simplicity and efficiency. Through meticulous training and evaluation, we sought to evaluate the efficacy and limitations of distillation in effectively capturing the teacher's knowledge.

Subsequent refinements introduced alterations to the teacher model aimed at optimising training dynamics and enhancing overall performance without compromising its feature extraction capabilities. By adjusting dropout rates and fine-tuning distillation parameters, we aimed to strike a delicate balance between learning from the teacher model and adhering to ground truth. These adjustments, coupled with structured loss functions and updates to the Distiller class, yielded promising outcomes. Post-distillation evaluation revealed improved validation performance of the distilled simpler student model compared to both the complex teacher model and the student model which has been trained from scratch. These observations underscore the success of our refined approach in achieving successful knowledge transfer and enhancing model generalisation.

In our pursuit of narrowing the accuracy gap between the teacher model and the distilled student model, we incorporated ResNet50, a pre-trained convolutional neural network renowned for its depth and regularisation capabilities, as the teacher model in our distillation process. While maintaining the student architecture largely unchanged, we strategically reduced dropout layers to enhance generalisation. This shift in methodology yielded substantial improvements, with the distilled student model achieving a satisfactory accuracy surpassing previous benchmarks.

By leveraging the pre-trained features and regularisation mechanisms inherent in ResNet50, we successfully addressed the widening accuracy gap observed in prior implementations, further advancing our research goals of achieving comparable accuracies and enhancing performance through distillation.

CONCLUSION AND FUTURE DIRECTION

The experimentation with various distillation techniques for plant disease detection highlighted their effectiveness in compressing model size and improving performance. Techniques like self-distillation and multi-teacher-student distillation demonstrated success in knowledge transfer, leveraging teacher model predictions to create compact yet accurate student models. However, challenges were encountered with certain methods, emphasising the need for nuanced understanding and selection of techniques based on dataset characteristics.

Our analysis showed that while some distillation methods, like multi-teacher-student distillation, achieved competitive accuracy, others fell short of the teacher model's performance. These findings stress the importance of comprehensive evaluation when choosing distillation techniques for practical applications.

Despite the challenges faced, distillation proved resilient for model compression and knowledge transfer, indicating its potential for improving agricultural practices. Opportunities for future research lie in optimising distillation techniques, integrating domain knowledge, and validating model effectiveness in diverse agricultural settings. Collaboration between machine learning experts and domain specialists could facilitate the development of context-aware and robust models.

Future Directions

To further improve the accuracies of our distilled models, several technical modifications can be explored. Adjusting distillation parameters like temperature and alpha can fine-tune the knowledge transfer process between teacher and student models. Modifying the student model's architecture by adjusting its depth, width, or adding skip connections can enhance its capacity to capture complex patterns. Incorporating regularisation techniques such as dropout and weight decay can prevent overfitting and promote generalisation. Experimenting with different optimization algorithms and learning rate schedules can optimise training convergence and final accuracy. Additionally, data augmentation techniques help to artificially diversify the training dataset, leading to enhanced generalisation.

Ensemble methods, like combining predictions from multiple student models or using multi-teacher distillation, can further reduce variance and improve accuracy. Attention mechanisms and transfer learning can also be leveraged to focus on informative features and utilise knowledge from pre-trained models, respectively. Finally, systematic hyperparameter tuning and fine-tuning on related tasks can help in finding the optimal configuration for our specific application, leading to improved performance in plant disease detection.

Future research should focus on optimising distillation techniques, incorporating domain-specific knowledge, and validating model effectiveness in real-world scenarios. Additionally, ethical and societal considerations must be addressed to ensure responsible deployment. By addressing these aspects, we can harness the potential of distillation to innovate and sustainably transform agriculture for future generations.

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